Impulse: Recall that we defined the idea of impulse while studying laws of motion. By second law, $\vec{F} = d\vec{p}/dt$ thus, $\Delta\vec{p} = \int \vec{F}\,dt$. The time integral of force is called impulse. Mathematically:

$$\vec{J} = \int \vec{F}\,dt$$

Impulse Momentum Theorem: Net impulse acting on a particle is equal to net change in it linear momentum. Mathematically, $\vec{J} = \Delta\vec{p}$. Now since this is a vector equation, it can also be applied component wise. Thus in terms of components:

$$J_x = \Delta p_x \,, \quad J_y = \Delta p_y \text{ and } J_z = \Delta p_z$$

Impulse Momentum Theorem For Systems: Net change in linear momentum of a system is equal to net impulse due to external forces. This is because all internal forces are action and reaction pairs thus upon summation over the entire system, the internal impulses also add up to zero. Thus:

$$\Sigma \vec{J}_i = \Sigma \Delta\vec{p}_i = \Sigma \vec{J}_{ext_i} \quad \text{thus:} \quad \boxed{\vec{J}_{ext} = \Delta\vec{p}_{sys}}$$

→ This equation is also a vector equation so can be applied component-wise. Thus:

$$\boxed{J_{ext_x} = \Delta p_{sys_x}} \quad \text{etc.}$$

Impulses During a Sudden Phenomena: During sudden phenomena like explosion, collision or sudden tightening of strings, the impulses of tension, normal reaction and explosive forces are very very large in comparison to the impulse of gravity during the small time window beginning just before the phenomenon and ending just after the phenomenon. Similarly if spring forces are involved during such phenomena, spring forces are also significantly smaller than the impulsive forces of tension, normal reaction and explosive forces, thus while analysing the impulsive phenomena, the impulses

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of relatively smaller forces like the gravity and spring forces are omitted.

→ While analysing impulsive phenomena in small time windows, the impulse of relatively smaller forces like gravity and spring forces are ignored.

FREE IMPULSE DIAGRAMS: Just as we draw free body diagrams for purpose of applying Newton's Second law in $\vec{F} = m\vec{a}$ form similarly for applying the Newton's law in the form $\vec{J}_{ext} = \Delta\vec{p}_{sys}$ we draw the free impulse diagrams and as mentioned earlier, if we are analysing a small time window phenomenon, we omit the impulse of relatively smaller forces of gravity and spring forces.

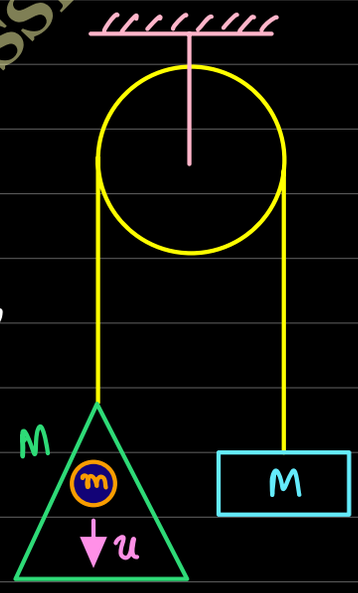
EXAMPLE: A dough ball of mass m collides with a pan of mass M which is counterbalanced by a block as shown.

(a) Draw the FID for each massive body

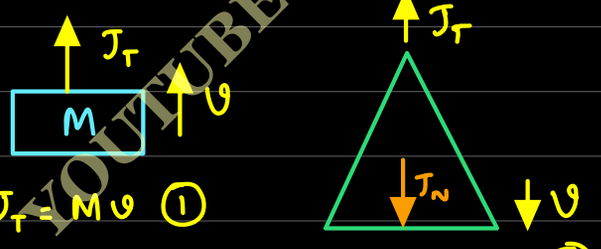
(b) Write Impulse momentum equation for each body

(c) Find post impulse speed of the system.

(Pulleys and strings are ideal).



Solution:

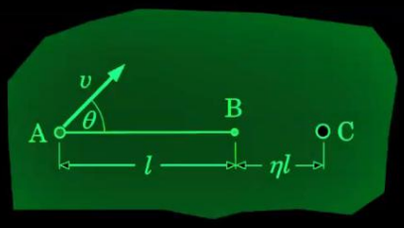




Considering the FID's we get the equations (1), (2) & (3) and solving, we get $v = mu/(2M+m)$ Ans.

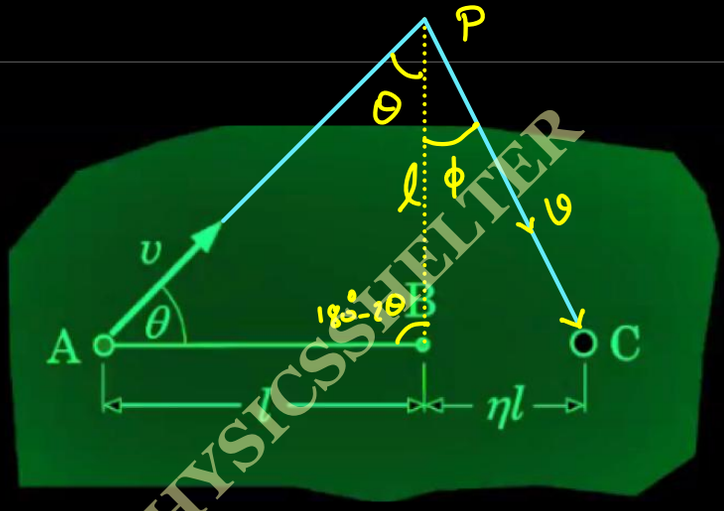
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Example: Pathfinder: Check 1:



A small disc A placed on a frictionless tabletop is connected to a small protrusion B on the tabletop with the help of an almost inextensible elastic cord of relaxed length l. There is a trap C in the table at distance $\eta = 0.5$ times of l from the protrusion. Initially the disc, the protrusion and the trap are in a line and the cord is straight and relaxed as shown in the top view of the setup. At what angle with the line AC must the disc be projected along the tabletop so that it enters the trap? Diameter of the protrusion is negligible as compared to the length of the thread.

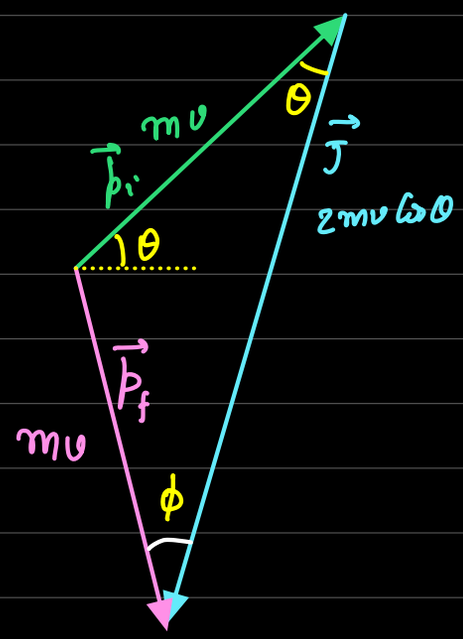
Solution: Elastic cord means that it does not cause any loss of mechanical energy. When the ball reaches point P, due to impulse its component of velocity along the string reverses while tangential component remains the same. Thus impulse is $2mv\cos\theta$ along PB. Now $\vec{p_i} + \vec{J} = \vec{p_f}$ and final momentum is directed along $\vec{PC}$. This gives us the vector diagram shown in adjacent figure. This is an isosceles triangle, thus, $\theta = \phi$.



Now applying Sine rule to $\triangle PBC$:

$$\frac{\eta l}{\sin\phi} = \frac{l}{\sin(180^\circ - (2\theta + \phi))}$$

$$\Rightarrow \frac{\eta l}{\sin\theta} = \frac{l}{\sin 3\theta} \Rightarrow \frac{\sin 3\theta}{\sin\theta} = \frac{1}{\eta}$$

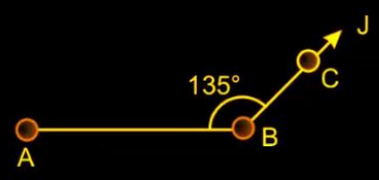


$$\Rightarrow \frac{3\sin\theta - 4\sin^3\theta}{\sin\theta} = \frac{1}{\eta} \Rightarrow 3 - 4\sin^2\theta = \frac{1}{\eta} \Rightarrow 4\cos^2\theta = \frac{1}{\eta} + 1$$

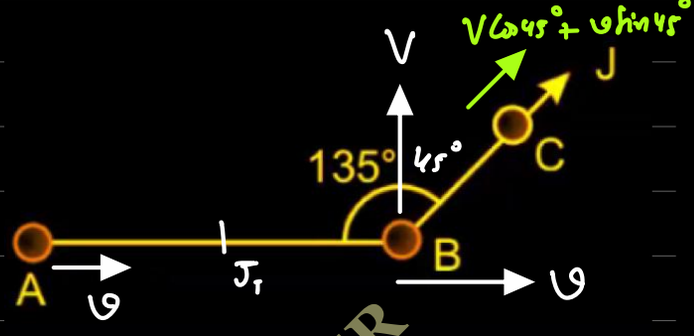
$$\Rightarrow \cos\theta = \sqrt{\frac{1+\eta}{4\eta}} = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ \text{ ans.}$$

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EXAMPLE: Three identical particles A, B and C lie on a smooth horizontal table. Light inextensible strings which are just taut connect AB and BC and $\angle ABC$ is $135°$. An impulse J is applied to the particle C in the direction BC. Find the initial speed of each particle. The mass of each particle is m. (Before impulse velocities are zero)



Solution: Since AB will become taut, the component of velocity of A along the string is same as that of B. Let this be v, and let component of velocity of A $\perp$ to AB be V. Now applying LIM (Linear Impulse Momentum theorem) along AB for the whole system:

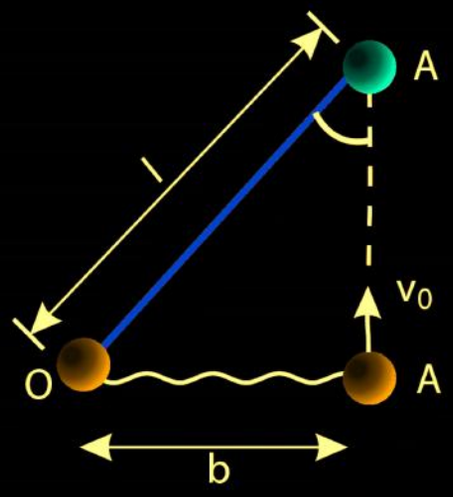


$$J\cos 45° = mv + mv + m\left(\frac{V+v}{\sqrt{2}}\right)\times\frac{1}{\sqrt{2}} \quad (1)$$

Now for $(B+C)$, $J\sin 45° = mV + m\left(\frac{V+v}{\sqrt{2}}\right)\cdot\frac{1}{\sqrt{2}}$ (2)

Now we can solve for v & V using (1) & (2).

EXAMPLE: Two balls of mass m each are connected by a relaxed string on a smooth table as shown. Now Ball A is given a velocity v_0. When the string becomes taut, find (a) Impulsive tension (b) Angular velocity of the system (c) Tension after the string becomes taut.

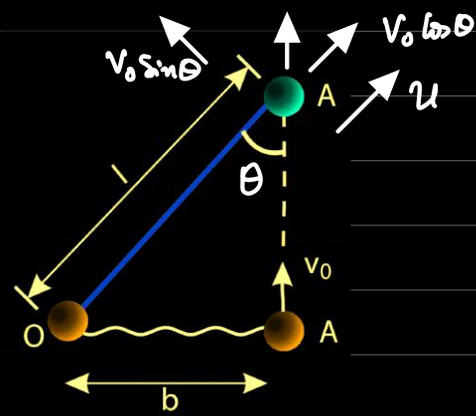


Solution: Since impulse of tension is internal, momentum of system remains conserved. Let final speed of the balls along the string be u then by COM along the string:

$$mv_0\cos\theta = 2mu \Rightarrow u = v_0\cos\theta/2.$$

Now applying impulse momentum theorem on O,

$$J_T = mu = \frac{mv_0\cos\theta}{2} = \frac{mv_0}{2}\frac{\sqrt{l^2-b^2}}{l} \text{ Ans.}$$



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Now since impulse on ball A is // to the string, momentum ⊥ to string remains conserved for A, hence its velocity ⊥ to string remains $v_0 \sin\theta$ and accordingly $\omega = \frac{v_0 \sin\theta}{\ell} = \frac{v_0 b}{\ell^2}$ Ans.

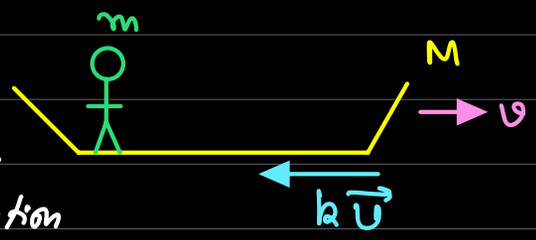
Now from CM frame, both balls execute circular motion about common CM in a radius $\ell/2$. Thus balancing forces in centrifugal direction: $m\omega^2 \frac{\ell}{2} = T \Rightarrow T = \frac{m}{2} \frac{v_0^2 b^2}{\ell^3}$ Ans.

EXAMPLE: PATHFINDER: MCQ 3:

A boy of mass 50 kg is standing at one end of a 9.0 m long boat of mass 100 kg that is floating motionless on a calm lake. The man walks to the other end of the boat and stops there. The boat also moves and finally stops due to water resistance. If force of water resistance is proportional to velocity of the boat relative to the water. What is the magnitude of the net displacement s of the boat?

(a) $s = 0.0$ m
(b) $s = 3.0$ m
(c) $s < 3.0$ m
(d) $s > 3.0$ m

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Solution: The initial momentum of the system is zero and so also the final momentum. Now applying $\vec{F}_{ext} = \Delta\vec{p}$ in horizontal direction for boy + boat system:

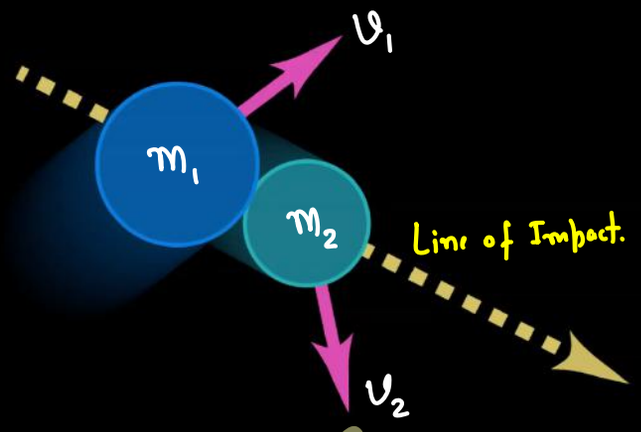
$$\int -k\vec{v}\,dt = 0 \Rightarrow -k\vec{s} = 0$$

thus $\vec{s} = 0$ Ans !!! ☺

COLLISIONS: Whenever two surfaces moving with different velocities suddenly come in contact, we have a collision. The collision forces the points of contact of two surfaces to move with same speed until the contact lasts. In this chapter we will be mainly dealing with collisions of smooth objects, where there is no impulse of friction during collisions. (Mostly collisions of

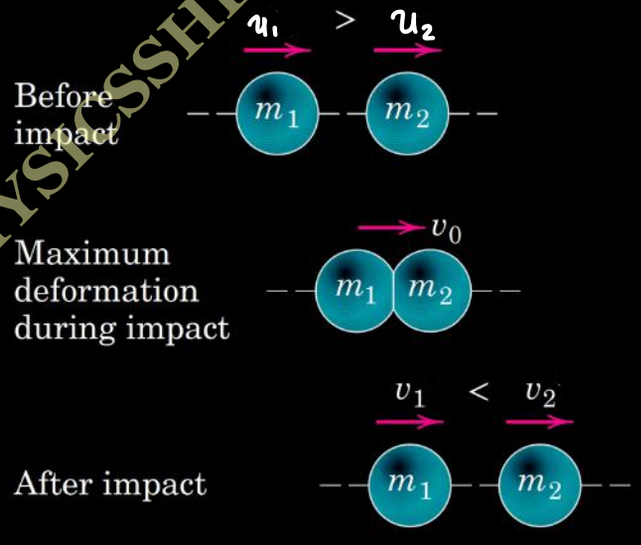
spheres where impulse of Normal reaction will pass through centre of mass).

LINE OF IMPACT: The common normal at the point of impact is known as line of impact (for collision of smooth spheres, this is the line joining centres of the spheres). This is also the line along which the impulse of normal reaction acts on the colliding bodies.



DIRECT COLLISIONS (HEAD-ON COLLISIONS):

When the velocities of the colliding bodies just before the collision are directed along line of impact, we call the collision head-on collision or direct collision. During the collision process, the surfaces of the colliding bodies deform and again reform (may not reform completely). All the while, normal reaction acts between them, thus causing change in velocities of the colliding bodies.

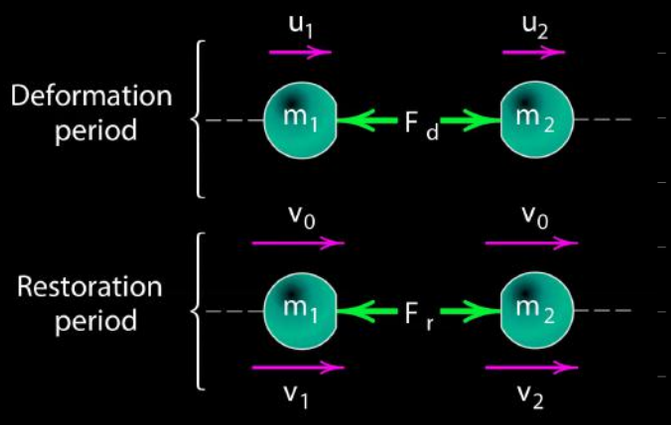


FREE COLLISIONS: If during collision of two bodies, there is no third body applying impulse on them, we call it a free collision, during free collisions, momentum of the system of colliding particles remains conserved. Thus in the figure shown above,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad (1)$$

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Restitution: Let F_r and F_d represent force of interaction between two colliding bodies during reformation and deformation respectively, then we define coefficient of restitution of the collision as the ratio of impulse of reforming force to that of the deformation force. Mathematically:



$$e = \frac{J_R}{J_D} = \frac{\int F_r\,dt}{\int F_d\,dt}$$ Now recall that from CM frame, the momenta of each colliding particle is μu_{rel} before collision and μv_{rel} after collision. Also since centre of mass is a zero momentum frame, at the time of maximum deformation, the momenta of individual bodies must be zero. Now let us focus on body 2. Before collision, its momentum is $\mu(u_2 - u_1)$, at maximum deformation, its momentum is zero. Thus by impulse momentum theorem, The impulse of deformation, $J_D = 0 - \mu(u_2 - u_1) = \mu(u_1 - u_2)$ Now after the collision is completed, its momentum is $\mu(v_2 - v_1)$. Thus, the impulse of reformation, $J_R = \mu(v_2 - v_1) - 0 = \mu(v_2 - v_1)$. Thus, by definition of coefficient of restitution,

$$e = \frac{J_R}{J_D} = \frac{\mu(v_2 - v_1)}{\mu(u_1 - u_2)} = \frac{v_2 - v_1}{u_1 - u_2} \quad \text{or} \quad \boxed{e = \frac{v_2 - v_1}{u_1 - u_2}}$$

→ Now recognize that $v_2 - v_1$ is velocity of separation just after the collision and $u_1 - u_2$ is velocity of approach before the collision. Thus for collisions, e can be alternatively defined as:

$$\boxed{e = \frac{\text{Velocity of Separation}}{\text{Velocity of Approach}}}$$

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Impulse Of Head-On Collision In Terms Of Restitution Coefficient:

Net impulse during collision is sum of the deformation and reformation impulses. Thus $J = J_R + J_D$ also as derived in a previous section, $J_D = \mu u_{rel}$ and $J_R = \mu v_{rel} = e J_D$ thus $J = \mu u_{rel} + e\mu u_{rel}$

Thus $\boxed{J = \mu u_{rel}(1+e)}$

→ This formula may be conveniently used for calculating post collision velocities using the impulse momentum theorem.

Energy Loss In Head-On Collisions In Terms Of Restitution Coefficient:

Recall the KE of a two particle system in CM frame is $\frac{1}{2}\mu v_{rel}^2$. Thus, the loss in KE during a head on collision as seen from CM frame is:

$$\Delta Q = \frac{1}{2}\mu u_{rel}^2 - \frac{1}{2}\mu v_{rel}^2 = \frac{1}{2}\mu u_{rel}^2 [1-e^2]$$

this loss of KE appears as heat, sound or bond deformations (strain energy). Since the work done by internal force pairs are independent of frame, therefore this liberated heat (or loss in KE) is also frame independent and happens due to internal forces of deformations. Thus Heat liberated:

$\boxed{\Delta Q = \frac{1}{2}\mu u_{rel}^2 [1-e^2]}$, fractional loss of energy, $\boxed{f = 1-e^2}$ (In CM frame)

Classification Of Collisions Based On Coefficient Of Restitution:

In the previous section we saw that energy lost in a collision is proportional to $1-e^2$, thus depending on the value of e, from CM frame, all energy may be lost, some energy may be lost or no loss of energy may occur. Thus we classify the collisions as:

Perfectly Elastic Collisions: The collisions in which mechanical energy is conserved ($e=1$) are called perfectly elastic collisions.

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PERFECTLY INELASTIC COLLISIONS: The collisions in which all the available KE in the CM frame is lost are called perfectly inelastic collisions. For such collisions, $e = 0$.

INELASTIC COLLISIONS: A collision in which there is some (or all) loss of mechanical energy is called an inelastic collision. Thus, for inelastic collisions, $0 \leq e < 1$ (from CM frame). Perfectly inelastic collisions are a special case of inelastic collisions with $e = 0$.

ANALYSIS OF HEAD ON COLLISIONS:

Recall that impulse during a head on collision is given as $J = \mu u_{rel}(1+e)$. Thus if before collisions, two spheres are moving with speeds u_1 and u_2, their post collision speeds are readily calculated as:



$$v_2 = u_2 + \frac{\mu(u_1 - u_2)(1+e)}{m_2} \quad (1)$$

and $$v_1 = u_1 - \frac{\mu(u_1 - u_2)(1+e)}{m_1} \quad (2)$$

Alternatively we could also find post collision speeds using momentum and restitution equations i.e: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ and $v_2 - v_1 = e(u_1 - u_2)$. Now depending on masses and coefficients of restitution, we have certain special cases worth memorizing:

PERFECTLY ELASTIC FREE HEAD-ON COLLISION BETWEEN IDENTICAL SPHERES:

Here, $m_1 = m_2 = m$, $e = 1$, $\mu = m/2$ thus by equation (1):

$$v_2 = u_2 + \frac{1}{2}[u_1 - u_2][2] = u_1 \quad \text{and by (2)}$$

$$v_1 = u_1 - \frac{1}{2}(u_1 - u_2)(2) = u_2 \quad \text{or} \quad v_2 = u_1 \text{ and } v_1 = u_2$$

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→ Thus we can say that in a perfectly elastic head-on collision between identical spheres, the velocities get exchanged. From dynamics point of view, it is as if the spheres have passed through each other.

Collision With A Very Massive Body:

Let $m_1 = m$ and $m_2 = M$ $(>> m)$. Then, $\mu = \frac{Mm}{M+m} \approx m$

thus using equation ① and considering $m/M \approx 0$, $v_2 = u_2$

→ Velocity of the massive body remains unchanged.

Now $\boxed{J = \mu u_{rel}(ite) \approx m u_{rel}(ite)}$. Now using ②,

$$v_1 = u_1 - (u_1 - u_2)(ite).$$

→ Note that here, the final speeds are nearly independent of mass of the very heavy body i.e final outcome will be nearly same for $M/m = 100$ or $M/m = 10^5$.

→ Collisions with floors, walls or balls with massive vehicles typically fall in this category and we don't need to know the exact mass of the massive object for solving such problems. The only unknown here is the final speed of lighter body which is readily calculated using restitution equation.

Perfectly Inelastic Head-On Free Collision:

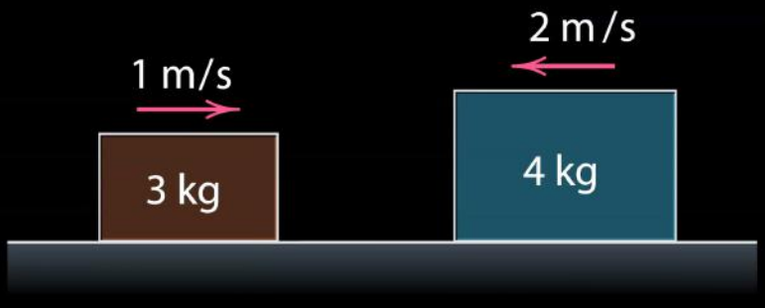
Here $e=0$ so $v_1 = v_2$, thus final common speed is readily calculated using momentum conservation: $m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$

thus:

$$\boxed{v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}}$$

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EXAMPLE: Two bodies have masses and velocities as shown in the figure. Coefficient of restitution, $e = 0.5$. If the duration of collision is 0.05 sec, calculate



(a) Deformation impulse (b) Reformation impulse (c) Total impulse (d) Post collision speeds (e) Average force of interaction.

Solution: $J_D = \mu(u_1 - u_2) = \frac{12}{7} \times 3 = 36/7$ kg m/s

$J_R = e\mu(u_1 - u_2) = eJ_D = 18/7$ kg m/s

$J = J_R + J_D = 54/7$ kg m/s.

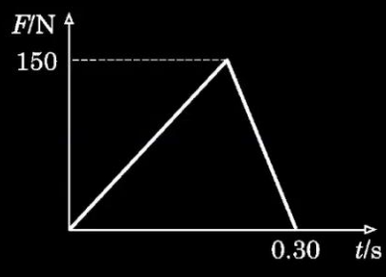
Now $V_4 = -2 + J/4 = -2 + \frac{54}{7} \times \frac{1}{4} = \frac{-1}{14}$ m/s Ans.

$V_3 = 1 - J/3 = 1 - \frac{54}{7} \times \frac{1}{3} = \frac{-11}{7}$ m/s Ans.

Now Average force $= \frac{1}{T}\int F dt = J/T = (54/7) \times (1/0.05) \approx 154$ N Ans.

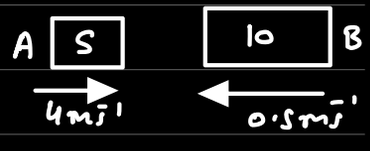
EXAMPLE: PATHFINDER: MCQ12:

Two bodies A and B of masses 5.00 kg and 10.0 kg moving in free space in opposite directions with velocities 4.00 m/s and 0.50 m/s respectively undergo a head-on collision. The force F of their mutual interaction varies with time t according to the given graph. What can you conclude from the given information?



(a) Period of deformation is 0.20 s.
(b) Coefficient of restitution is 0.5.
(c) Body A will move with velocity 0.5 m/s in the original direction.
(d) Body B will move with velocity 1.75 m/s in reverse direction.

Solution: Total Impulse $J = \frac{1}{2} \times 0.3 \times 150 = 22.5$ Ns.



Now impulse of deformation, $J_D = \mu(u_1 - u_2) = \frac{50}{15} \times (4.5) = 15$ Ns

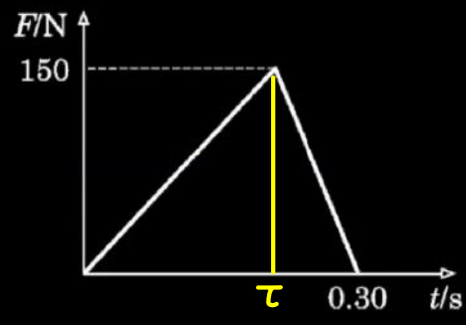
Thus impulse of reformation $J_R = J - J_D = 7.5$ Ns

Now $e = J_R / J_D = 0.5$ (b) is correct.

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Now let τ be the duration of deformation

impulse then, $\frac{1}{2} \times 150 \times \tau = 15 \Rightarrow \tau = 0.2$ sec



thus (a) is also correct

Now velocity of 5 Kg block: $v_5 = 4 - \frac{22.5}{5}$

$\Rightarrow v_5 = -0.5\ ms^{-1}$ i.e $0.5\ ms^{-1}$ in reverse direction. So (c) is wrong.

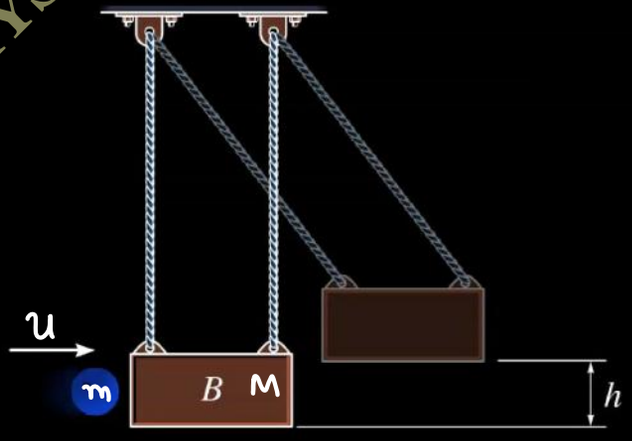
Now for 10 Kg block, $v_{10} = -0.5 + \frac{22.5}{10} = +1.75\ ms^{-1}$

thus 10 Kg reverses its direction and moves with $1.75\ ms^{-1}$, option d is correct.

thus correct options are (a), (b), (d).

EXAMPLE: BALLISTIC PENDULUM:

A bullet of mass m is fired in to a block of mass M and gets embedded instantaneously. Through what height does the block rise afterwards.



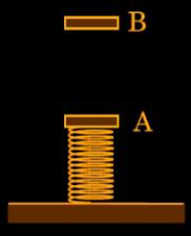
Solution: It is a case of perfectly inelastic collision. Thus velocity just after embedding $v = mu/(M+m)$.

Now by energy conservation: $\frac{1}{2}(M+m)v^2 = (M+m)gh \Rightarrow h = \frac{v^2}{2g}$

thus $h = \left(\frac{mu}{M+m}\right)^2 \cdot \frac{1}{2g}$ Ans.

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Example: Pathfinder: Build II:



A thin plate A of mass m is affixed on the upper end of a spring, lower end of which is affixed on the ground. In equilibrium, the spring is compressed by an amount x_0. Another thin plate B of mass $2m$ dropped from a height $3x_0$ above plate A hits the plate A, moves downwards together with the plate A and after reaching a lowest position, both the plates rebound upwards. What maximum height above the initial position of the plate A will the plate B rise?

Solution: It is a case of inelastic collision, thus loss in ME:

$$\text{Loss} = \frac{1}{2}\mu u_{n1}^2(1-e^2) = \frac{1}{2}\cdot\frac{2m}{3}\times\left(\sqrt{2g\times 3x_0}\right)^2 = 2mgx_0$$

Now for B to lose contact with A, pseudo force on it must be equal to its weight i.e acceleration of A must be g i.e the spring must be at natural length. Let V be the velocity at the time of contact loss, now applying post collision energy conservation till the moment of separation (For gravitational PE, initial level of A is considered as reference):
(Also recall that spring energy is $\frac{1}{2}F\times$ compression). Thus:

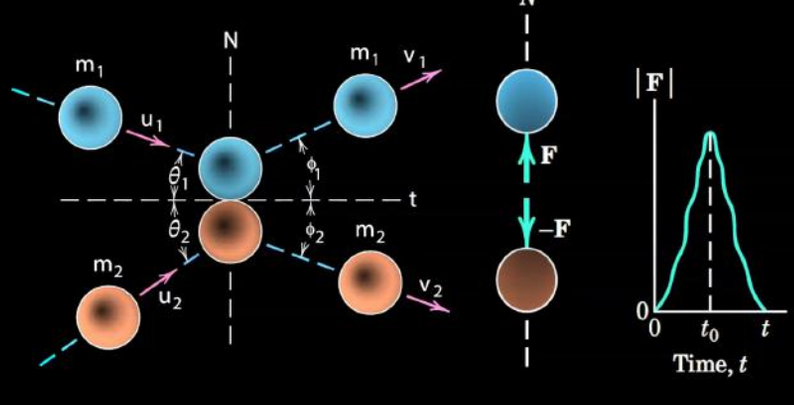
$$\underbrace{6mgx_0}_{B} - \underbrace{2mgx_0}_{\text{loss}} + \underbrace{\frac{1}{2}\cdot mgx_0}_{\text{Spring}} = \underbrace{\frac{1}{2}(2m+m)V^2}_{\text{Combined KE}} + \underbrace{(2m+m)gx_0}_{\text{Combined PE}}$$

Solving we get $V^2 = gx_0$, thus height risen from this position: $\frac{V^2}{2g} = \frac{x_0}{2}$

Thus total height from initial level of A, $x_0 + \frac{x_0}{2} = \frac{3}{2}x_0$ Ans.

Oblique Free Collisions:

When the velocity of atleast one of the colliding bodies is not along the line joining the centres of the spheres, we call the collision as indirect or oblique collision. Here we can resolve the velocities in to components



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along the line of impact and ⊥ to line of impact. The components ⊥ to line of impact remain unchanged. If we forget the components ⊥ to line of impact, the collision looks just like the head on collision and all equations remain the same except u_1 and u_2 get replaced by u_{1N} and u_{2N} and v_1 and v_2 get replaced by v_{1N} and v_{2N} where subscript N represents components along line of impact. Thus for such collisions:

→ $$e = \frac{v_{2N} - v_{1N}}{u_{1N} - u_{2N}}$$

→ $$J_D = \mu |u_{1N} - u_{2N}| \qquad J_R = \mu (v_{2N} - v_{1N}) = e J_D \qquad \text{and} \qquad J = (1+e)\mu u_{rel_N}$$

→ Loss of Mechanical Energy : $$\Delta Q = \frac{1}{2}\mu u_{rel_N}^2 [1 - e^2]$$

→ In perfectly elastic collisions between spheres of same mass velocity components along line of impact get exchanged

→ In perfectly inelastic collisions, the components of velocities along the line of impact become equal

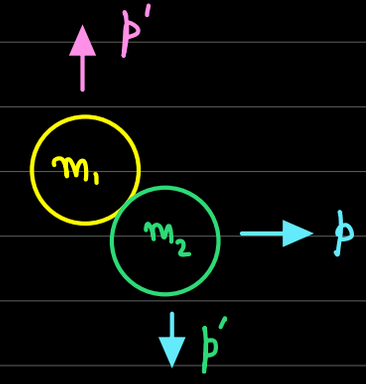
→ In a collision with a massive body, the velocity of the massive body remains unchanged and the velocity of the lighter body can be found using restitution equation.

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EXAMPLE : IRODOV 1·168 :

A particle of mass m_1 experienced a perfectly elastic collision with a stationary particle of mass m_2. What fraction of the kinetic energy does the striking particle lose, if
(a) it recoils at right angles to its original motion direction;
(b) the collision is a head-on one?

Solution: (a) Post collision momentum variables are as shown in adjacent figure (using momentum conservation considerations).



Now:

$$\frac{p^2}{2m_1} = \frac{p'^2}{2m_1} + \frac{p^2}{2m_2} + \frac{p^2}{2m_2}$$

$$\Rightarrow \frac{p'^2}{2\mu} = \frac{p^2}{2}\left[\frac{1}{m_1} - \frac{1}{m_2}\right]$$

Now KE $\propto p^2$ thus fractional energy lost $= 1 - \frac{p'^2}{p^2}$

Now $\frac{p'^2}{p^2} = \frac{m_1 m_2}{m_1 + m_2}\left[\frac{m_2 - m_1}{m_1 m_2}\right] = \frac{m_2 - m_1}{m_1 + m_2}$

$\Rightarrow$ Fraction of energy lost, $1 - \frac{p''^2}{p^2} = \frac{2m_1}{m_1 + m_2}$ Ans.

(b) In head on collision, $J = 2\mu u$ (if u is incidence speed)

$\Rightarrow$ momentum of recoiling particle : $p' = mu - 2\mu u$

$\Rightarrow p'/p = \frac{(m - 2\mu)}{m}$

$$= 1 - \frac{2m_2}{m_1 + m_2} = \frac{m_1 - m_2}{m_1 + m_2}$$

$\Rightarrow$ lost fraction $\eta = 1 - \left(\frac{p'}{p}\right)^2 = 1 - \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 = \frac{4m_1 m_2}{(m_1 + m_2)^2}$ Ans.

EXAMPLE : IRODOV 1.169:

Particle 1 experiences a perfectly elastic collision with a stationary particle 2. Determine their mass ratio, if

(a) after a head-on collision the particles fly apart in the opposite directions with equal velocities;

(b) the particles fly apart symmetrically relative to the initial motion direction of particle 1 with the angle of divergence $\theta = 60°$.

Solution: (a) $J = 2\mu u \Rightarrow m_1 u - 2\mu u = -m_1 v$ (1) ($\because$ reversal)

and, $2\mu u = m_2 v$ (2)

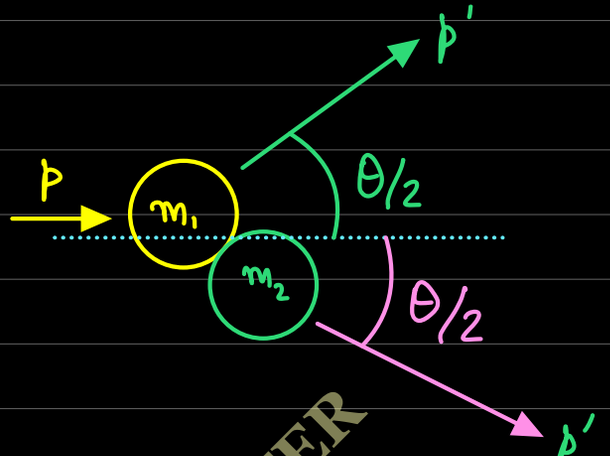
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$$\textcircled{1}/\textcircled{2} \Rightarrow : \quad \frac{m_1 - 2\mu}{2\mu} = -\frac{m_1}{m_2} \quad \Rightarrow \quad \frac{m_1+m_2}{2m_2} = 1 - \frac{m_1}{m_2}$$

$$\Rightarrow \frac{m_1+m_2}{2} = m_2 - m_1 \Rightarrow m_1 + m_2 = 2m_2 - 2m_1 \Rightarrow 3m_1 = m_2 \Rightarrow$$

$m_1/m_2 = 1/3$ Ans.

(b) By momentum conservation considerations post collision momenta are as shown.



Applying COM along initial line:

$$2p'\cos\theta/2 = p \quad \textcircled{1}$$

COME: $$\frac{p'^2}{2m_1} + \frac{p'^2}{2m_2} = \frac{p^2}{2m_1} \quad \textcircled{2}$$

Using $\textcircled{1}$ in $\textcircled{2}$: $$\frac{p'^2}{2\mu} = \frac{4p'^2\cos^2(\theta/2)}{2m_1} \Rightarrow \frac{m_1+m_2}{m_2} = 4\cos^2(\theta/2)$$

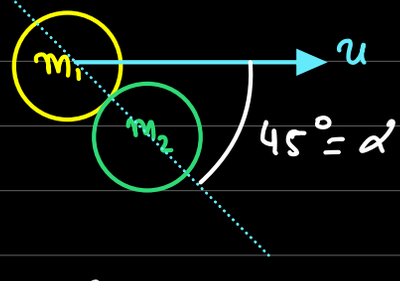
$$\Rightarrow \frac{m_1}{m_2} = 4\cos^2(\theta/2) - 1 = 4\left(\frac{1+\cos\theta}{2}\right) - 1 = 2\cos\theta + 1 = 2 \text{ Ans.}$$

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EXAMPLE : IRODOV 1.170:

A ball moving translationally collides elastically with another, stationary, ball of the same mass. At the moment of impact the angle between the straight line passing through the centres of the balls and the direction of the initial motion of the striking ball is equal to $\alpha = 45°$. Assuming the balls to be smooth, find the fraction η of the kinetic energy of the striking ball that turned into potential energy at the moment of the maximum deformation.

Solution: At maximum compression, loss in KE from CM frame must go in to potential energy thus, $\Delta U = \frac{1}{2}\mu u_{rel}^2 = \frac{1}{2}\mu(u\cos\alpha)^2$



Now fraction converted : $\eta = \Delta U/(\frac{1}{2}m_1u^2) = \frac{\mu}{m_1}\cos^2\alpha$

thus $\eta = \frac{m_2\cos^2\alpha}{(m_1+m_2)} = \frac{1}{2}\cos^2\alpha = \frac{1}{4}$ Ans $(\because m_1 = m_2)$.

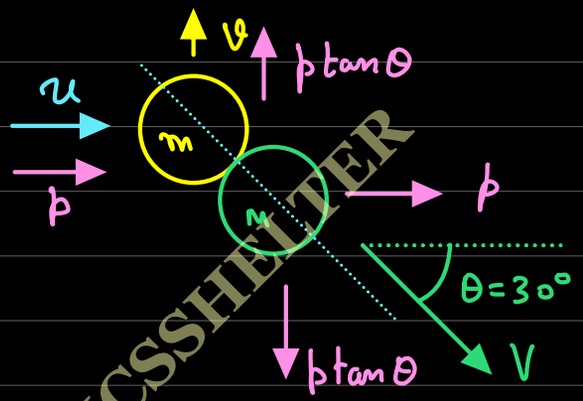
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Example: Irodov 1.173:

A particle of mass m having collided with a stationary particle of mass M deviated by an angle $\pi/2$ whereas the particle M recoiled at an angle $\theta = 30°$ to the direction of the initial motion of the particle m. How much (in per cent) and in what way has the kinetic energy of this system changed after the collision, if $M/m = 5.0$?

Solution: Let initial momentum of incident particle be p. Then by momentum conservation considerations, post collision momentum components are as shown in the figure.



Now, Initial KE, $KE_i = p^2/2m$

Final KE:

$$KE_f = \frac{p^2[1+\tan^2\theta]}{2M} + \frac{p^2\tan^2\theta}{2m}$$

$$\Rightarrow \frac{\Delta KE}{KE} = \frac{2m}{p^2}\left[\frac{p^2}{2m}[\tan^2\theta - 1] + \frac{p^2}{2M}[\sec^2\theta]\right]$$

$$= \tan^2\theta - 1 + \frac{m}{M}\sec^2\theta = \tan^2\theta - 1 + \frac{m}{M}(1+\tan^2\theta)$$

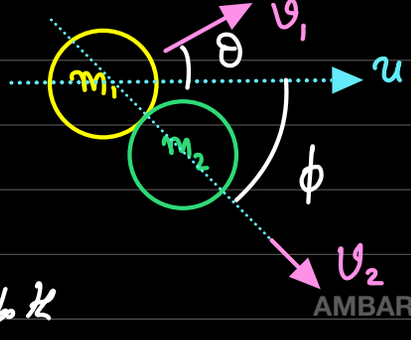
$$= \tan^2\theta[1 + m/M] + m/M - 1 = -40\% \quad \text{Ans.}$$

Example: Irodov 1.175:

A particle of mass m_1 collides elastically with a stationary particle of mass m_2 ($m_1 > m_2$). Find the maximum angle through which the striking particle may deviate as a result of the collision.

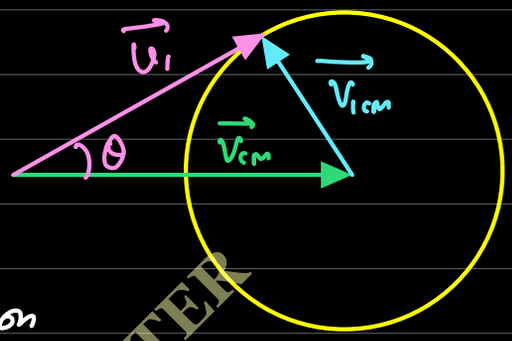
Solution: This problem is best analysed in CM frame. From CM frame momentum magnitude of each particle is $p_0 = \mu u$ both before and after the collision.

(One can readily see that this satisfies both



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energy and momentum conservation in the CM frame). Thus post collision speed of m_1, $U_{1cm} = \mu u / m_1$. The direction of the velocity will depend on impact parameter. ($\perp$ distance between line of u and centre of m_2). The net velocity of m_1 will be the resultant $\vec{V}_1 = \vec{V}_{cm} + \vec{V}_{1cm}$.

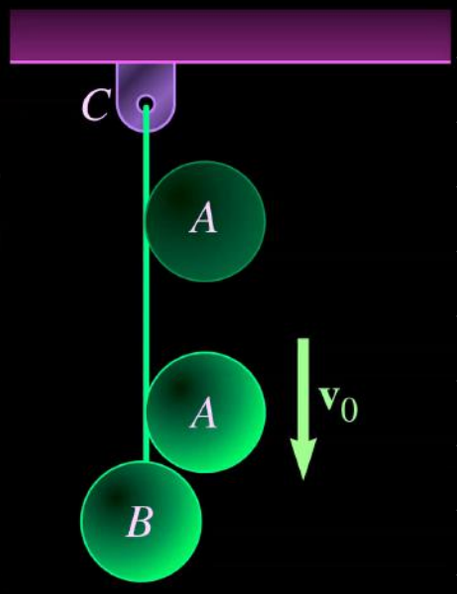


Now since magnitude of V_{1cm} remains constant, To represent all possible values of $\vec{V}_1$ we can draw a circle of radius $\vec{V}_{1cm}$ on head of $\vec{V}_{cm}$ and the resultant is from tail of $\vec{V}_{cm}$ to any point on the circle depending on chosen direction of $\vec{V}_{1cm}$. Refer to the figure, Here θ is the angle of scattering which is clearly maximize when $\vec{V}_1$ is a tangent to the circle. Thus, maximum deviation angle is given by $\sin\theta = V_{1cm} / V_{cm}$., Now $V_{cm} = m_1 u / (m_1 + m_2)$

and $U_{1cm} = \mu u / m_1 = m_2 u / (m_1 + m_2)$ thus $\boxed{\sin\theta = m_2 / m_1}$ Ans.

CONSTRAINED COLLISIONS:

When during a collision, the colliding bodies experience an impulse other than the impulse of their mutual impact, we will call the collisions constrained. For example consider the situation in adjacent figure. Here when ball A hits ball B, the ball B also experiences impulsive tension apart from impulse of normal reaction between the balls. For constrained collisions following equations are helpful:

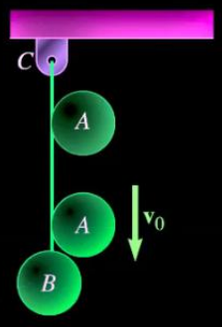


① Conservation of momentum in direction $\perp$ to "third party" impulse.
② Equation of restitution along line of impact
③ Impulse momentum equations for entire system
④ Impulse momentum equations for individual bodies.

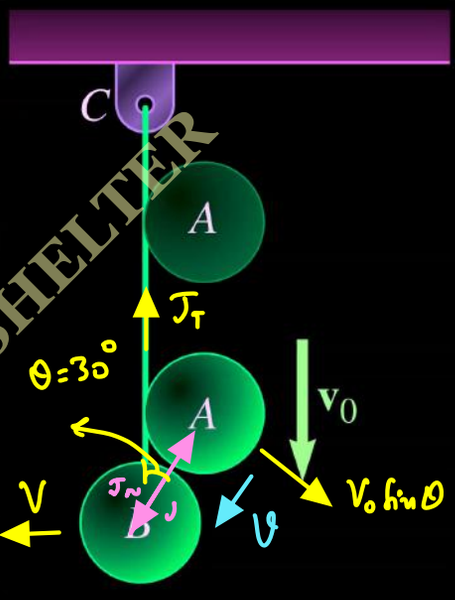
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Example:

Ball B is hanging from an inextensible cord BC. An identical ball A is released from rest when it is just touching the cord and acquires a velocity v_0 before striking ball B. Assuming a perfectly elastic impact ($e = 1$) and no friction, determine the velocity of each ball immediately after impact.
Mass of both balls is m each.



Solution: This is an oblique impact. The tangential velocity of A, i.e $v_0 \sin\theta$ remains unchanged during impact (∵ there is no impulse on A in that direction). Let post collision velocity of A be v along the normal and that of B be V along horizontal as shown then:



Conservation of momentum in x direction gives:

$$0 = mV + m(v\sin\theta - v_0\sin\theta\cos\theta) \quad (1)$$

(Remember to consider velocity of A carefully).

Now restitution equation gives:

$$V\sin\theta - v = v_0\cos\theta \quad (2)$$ Now solving (1) & (2),

$$V = \frac{2v_0\sin\theta\cos\theta}{1+\sin^2\theta}$$ Ans and $$v = \frac{-v_0\cos^2\theta}{1+\sin^2\theta}$$

Now speed of ball A is found as $v_A = \sqrt{v^2 + v_0^2\sin^2\theta}$

Think: How will you find J_T?

Solution: Using Impulse momentum theorem for both balls combined:

$$J_T = m(v_0 - (v\cos\theta + v_0\sin^2\theta))$$

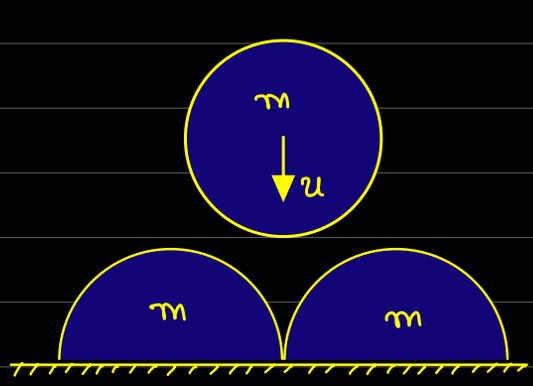
Think: How will you find J_N?

Solution: Using Impulse momentum theorem for ball B:

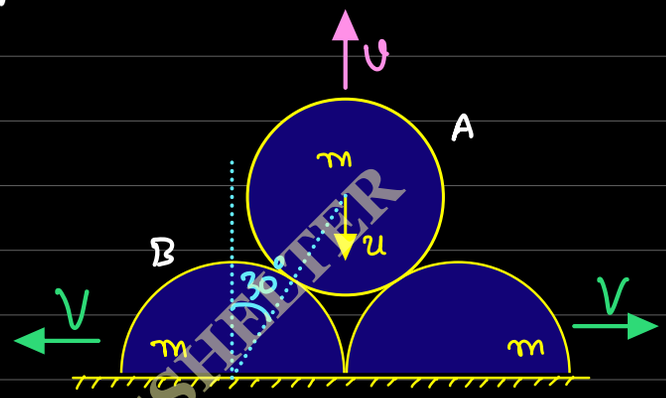
$$J_N\sin\theta = mV \Rightarrow J_N = mV/\sin\theta.$$

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EXAMPLE: Two smooth hemispheres of radius R and mass m are lying on a frictionless horizontal surface. They are hit symmetrically by a sphere of mass m and radius R. If the collision is perfectly elastic, what are the post collision speeds of the hemispheres.



Solution: Post collision speeds are shown in the adjacent diagram. Making restitution equation between bodies A & B:



$$V\cos 60° + v\cos 30° = u\cos 30° \quad (1)$$

Now let impulse of Normal reaction between A and B be J_N, Now applying Impulse momentum equation to B:

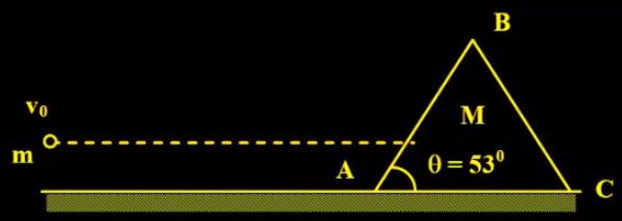
$$J_N \sin 30° = mV \quad (2)$$ Similarly Impulse momentum equation for A:

$$2J_N \cos 30° = mv + mu \quad (3)$$

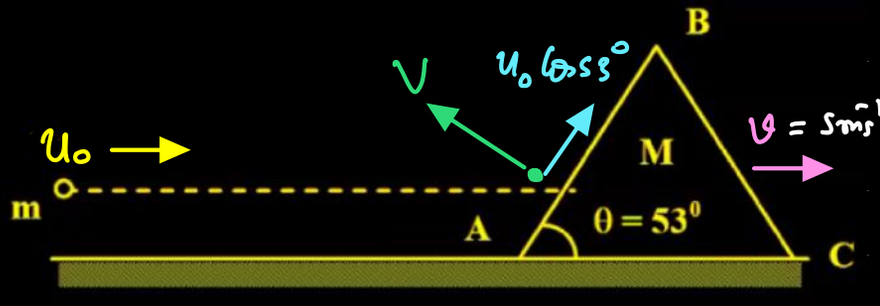
Now Solving (1), (2) & (3): $v = \frac{5u}{7}$, $V = \frac{2\sqrt{3}u}{7}$, $J_N = \frac{4\sqrt{3}}{7}mu$ Ans.

EXAMPLE:

A wedge ABC of mass M = 0.8 kg rests on a smooth surface. A shell of mass m = 0.1 kg moving horizontally with velocity u_0 = 40 m/s collides with the wedge. Find the maximum height reached by the shell if after collision the wedge starts moving towards right with velocity v = 5 m/s. Also find the coefficient of restitution.



Solution: Let component of velocity ⊥ to wedge after collision be V, by conservation of momentum in x direction:



$$mu_0 = Mv + m(u_0\cos 53° - V\cos 37°) \quad (1)$$

Solving we get $V = \frac{25Mv - 16mu_0}{20m} = 18\,ms^{-1}$.

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Now vertical speed after collision: $u_0 \cos 53^\circ \cos 37^\circ + V \cos 53^\circ = 30\ ms^{-1}$

Thus maximum height $= 30^2/2g = 900/20 = 45\ m$ Ans.

Now $e = \dfrac{V + v \sin 53^\circ}{u_0 \sin 53^\circ} = \dfrac{11}{16}$ Ans.

Restitution Coefficient For Elastic Strings:



Consider two blocks A and B connected by an extensible string, if now A were to be given some speed, from CM frame, the blocks will move apart and then come to momentary rest. The impulse of tension till this moment is called Impulse of deformation (J_D), Now string will try to contract again until it becomes loose once again, the Impulse of tension during this phase is impulse of reformation (J_R). The ratio of impulse of reformation to that of deformation is called coefficient of restitution. Thus,

$$e = \frac{J_R}{J_D} = \frac{\text{Velocity of Approach}}{\text{Velocity of Separation}}$$ ← See carefully!!

[In strings, first there will be separation (before tightening) and then there will be approach, unlike collisions).

Example: Pathfinder: Build 18:

Two blocks A and B of masses m_A and m_B placed on a frictionless horizontal floor are connected by a light and almost inextensible cord. The block A is given a velocity u away from the block B. If coefficient of restitution defined for the cord has a value e, find velocity with which block B begins to move.



Solution: $J = (1+e)\mu V_{rel} = (1+e)\mu u$, thus velocity of $B = J/m_B$

i.e. $V_B = \dfrac{(1+e)\mu}{m_B} u = \dfrac{(1+e)m_A}{m_A + m_B} u$ Ans.

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Example: Pathfinder: MCQ11:

A block of mass m is given a velocity towards another block of mass M kept at rest on a frictionless horizontal floor some distance apart from a wall as shown in the figure. If all the collisions are elastic, under which of the following conditions, will the blocks collide only once?



(a) $m \le M$ (b) $M \le 3m$
(c) $M \ge 3m$ (d) $M \le 3m \le 3M$

Solution: The post collision speed of m must not be more than that of M (speed could be directed either forward or backward). (This is because after collision with wall, velocity will simply reverse).

Now impulse: $J = 2\mu u$ where u is speed imparted to m.

Thus, speed of M:

$$v_M = 2\mu u/M$$

and that of m: $v_m = |u - 2\mu u/m|$ thus we want

$$|u - 2\mu u/m| \le (2\mu u/M) \text{ or } \left|1 - \frac{2M}{M+m}\right| \le \frac{2m}{M+m}$$

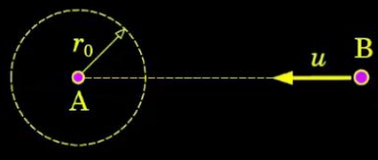
$$\Rightarrow \left|\frac{m-M}{M+m}\right| \le \frac{2m}{M+m} \text{ or } |m-M| \le 2m, \text{ now if}$$

$M \le m$, this is automatically satisfied. or else if $M > m$ then,

$M - m \le 2m \Rightarrow M \le 3m$ combining the two solution branches, $M \le 3m$ Ans. [option (b)]

Example: Pathfinder: Build-7:

Two particles A and B repel each other by a force $F = 6.0$ N when distance between them is less than $r_0 = 1.0$ m and apply no force on each other when distance between them is greater than r_0. Mass of particle A is $m = 1.0$ kg and mass of particle B is $\eta = 3$ times the mass of A. If the particle B is projected with velocity $u = 2.0$ m/s towards the particle A kept at rest in free space, find the minimum distance between them.



Solution: Recognise that the force of repulsion between two particles forms an action reaction pair. Also recall from the chapter on work power and energy that work done by any action reaction pair is independent of the

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translating frame.

Thus we might as well consider the work done from frame of one of the particles. From the centre of mass frame, at the nearest distance kinetic energy of the system will be zero (∵ Head on collision type of situation). Thus work done by force must cancel out initial KE in the Centre of mass frame. Now let nearest distance between them be r. Then by work energy theorem:

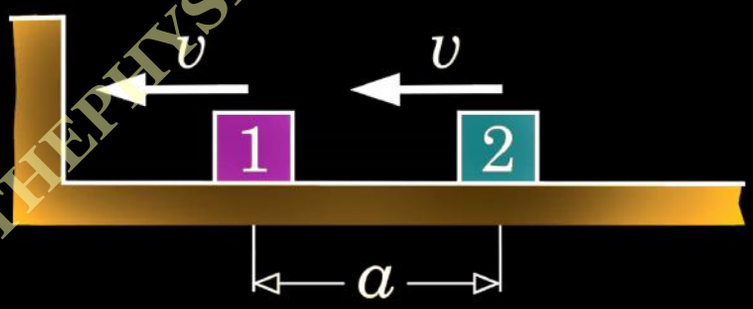
$$F(r_0 - r) = \frac{1}{2}\mu u_{rel}^2 \Rightarrow r = r_0 - \frac{1}{2}\frac{\mu u^2}{F} = r_0 - \frac{\eta m u^2}{2(\eta+1)F} \quad \text{Ans.}$$

$$\Rightarrow \quad r = 1 - \frac{1}{2} \times \frac{3}{4} \times 1 \times \frac{2^2}{6} = 3/4 \text{ m} \quad \text{Ans.}$$

(The Force is a central force from CM so we could also define $U(r)$ between particles).

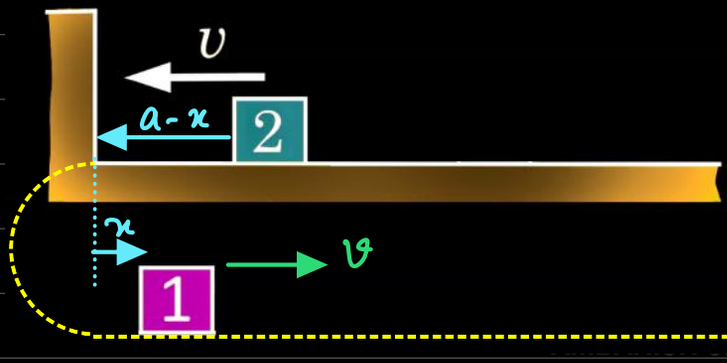
EXAMPLE: PATHFINDER BUILD 21: SIMPLIFIED:

Two identical blocks moving on a perfectly smooth surface are approaching a wall. If all collisions are perfectly elastic find the final speeds of the blocks and their separation. All collisions are elastic.



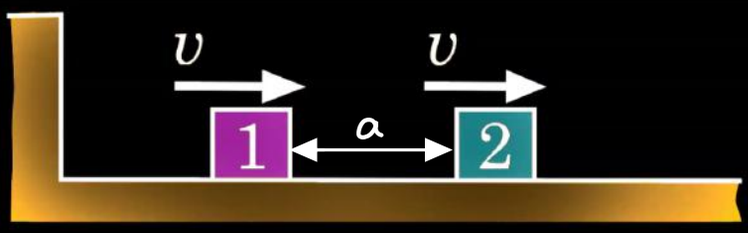
Solution: When a block collides with a wall elastically, its velocity reverses. Also whenever two blocks collide elastically, their velocities get exchanged, this is as if the blocks have passed through each other (unless they have some distinguishing mark on them.). The velocities exchange but not the blocks. This makes us collision process like a train taking a U turn but keeping its coaches in the original order.

The argument will be clearer from the adjacent graphic. Block 1 has taken the U turns and now velocities of 1 & 2 are about to be exchanged. The process looks like train taking a U turn.



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Once the "train" has completed the U turn, the separation between coaches is restored but the order of coaches remains the same. Thus final separation between the blocks remains a with velocities reversed. Thus final situation looks as shown in the figure. After completion of all collisions, the process looks as if the train has started moving in the reverse gear. The same reversal of train idea can be extended to a train of multiple blocks (for example pathfinder build 21 actual problem, in which we have a train of four blocks).

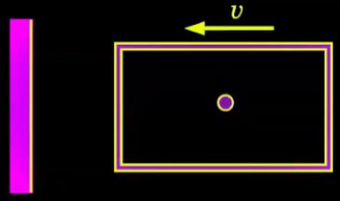


Example: Pathfinder: Build-19:

In gravity free space, a small ball is placed at the centre of a box of the same mass. The ball and the box both are moving with a velocity v towards a wall. Assume all collisions to be perfectly elastic.

(a) How many collisions are possible?

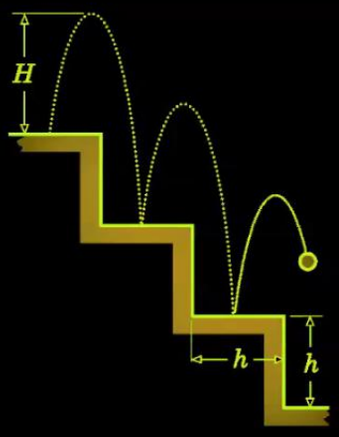
(b) Find velocities of the box and the ball and position of the ball relative to the box after all possible collisions.



Solution: If we think carefully, this problem is exactly the same problem as the previous one in disguise. We can think of leading edge of the box as block 1 and the ball as block 2. As the train reverses, the relative position of the ball and the box remain the same and both are travelling in the reverse direction. The total number of collisions can also be counted by analogy from previous problem. The collision sequence is: 1 — Wall, 1 → 2, 1 → Wall (each time a coach turns around the wall or coaches pass each other, there is a collision. Thus total number of collision: 3 Ans.
Final position of ball: mid point of the box. Final velocities of the ball and the box are v each in reverse direction Ans.

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Example: Pathfinder: MCQ : 25-27:



A marble is bouncing down a stairs in a regular manner, hitting each step at identical points and then rising the same height H above each step. Height and depth of a stair each are equal to h as shown in the figure. Coefficient of restitution for each bounce is e.

25. Air time between two successive bounces is

(a) $\sqrt{\frac{2h(1+e)}{g(1-e)}}$ (b) $\sqrt{\frac{2h(1-e)}{g(1+e)}}$

(c) $\sqrt{\frac{1.5h(1+e)}{g(1-e)}}$ (d) $\sqrt{\frac{1.5h(1-e)}{g(1+e)}}$

26. Horizontal component of velocity of the marble is

(a) $\sqrt{\frac{2gh(1-e)}{(1+e)}}$ (b) $\sqrt{\frac{2gh(1+e)}{(1-e)}}$

(c) $\sqrt{\frac{gh(1-e)}{2(1+e)}}$ (d) $\sqrt{\frac{gh(1+e)}{2(1-e)}}$

27. The height H attained by the marble after a bounce is

(a) $\frac{eh}{1-e}$ (b) $\frac{2eh}{1-e}$

(c) $\frac{2h}{1-e}$ (d) $\frac{e^2h}{1-e^2}$

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Solution: Normal Velocity just after a bounce $= \sqrt{2gH}$ and velocity just before a bounce: $\sqrt{2g(H+h)}$, Thus, $e = \sqrt{H/(H+h)}$ ①

Now air time can be found using $\vec{v} = \vec{u} + \vec{a}T$, thus,

$T = [\sqrt{2gH} + \sqrt{2g(H+h)}]/g$ ②, Now from ①, $H = e^2h/(1-e^2)$ ③

Using ③ in ②, $T = \sqrt{\frac{2h}{g}}\left[\frac{1+e}{\sqrt{1-e^2}}\right] = \sqrt{\frac{2h}{g}\left(\frac{1+e}{1-e}\right)}$ Ans. (a) [25]

[26] Now during this time of flight marble covers a horizontal distance h, thus horizontal speed $= h/T = \sqrt{\frac{gh}{2}\left(\frac{1-e}{1+e}\right)}$ Ans (c) [26]

[27] Now height attained by the marble was found in equation ③ as $H = e^2h/(1-e^2)$ which is option (d) Ans [27].

EXAMPLE; PATHFINDER: CHECK-7:

A small block of mass 490 g rests close to the left wall of a box of equal mass. Length of the box is 40 cm. The box is placed on a horizontal floor lubricated with oil. Force of viscous friction of the lubricating oil is given by equation $\vec{F} = -k\vec{v}$, where k = 0.5 kg/s and $\vec{v}$ is velocity of the box relative to the floor. There is no friction between the block and the floor of the box. If the block is given a velocity 10 m/s towards the right, how many times will the block collide with walls of the box and how far will the box move?



Solution: Recognize that mass of the box and the block are identical therefore at every collision, one of them comes to rest. For example initially the block travels a distance l and hits the box and comes to rest and then box's rear wall hits it again after travelling a distance l and then box comes to rest and block starts to move (starting a new cycle). Thus in one cycle box has moved a total distance l. In each cycle there are two collisions. Finally after the block hits the front wall of the box, the box will come to rest in infinite time but a finite distance (Thus last collision is on the front wall of the box). Thus total number of collisions will be odd. Thus if S is the total distance travelled by the box, the number of collisions will be $N = 2\,\text{GIF}\left[\frac{S}{l}\right] + 1$

Now it remains to find the value of S. For this purpose we will use Impulse Momentum Theorem for the system as a whole. Thus:

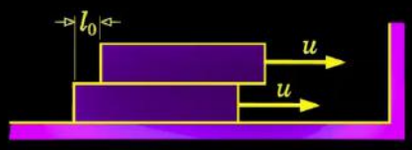
$\int kv\,dt = mu$ where m is the mass of the box and u initial speed. Thus $kS = mu$ or $S = mu/k$ Ans. Now putting in the values, $S = (0.49 \times 10/0.5)\,m = 9.8\,m$ Ans

Now $N = 2\,\text{GIF}(S/l) + 1 = 2\,\text{GIF}(9.8/0.4) + 1 = 49$ Ans.

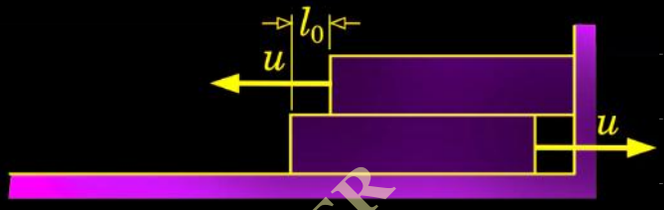
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EXAMPLE: PATHFINDER: CHECK-6:

A bar is placed with a slight offset l_0 on another identical bar that rests on a horizontal frictionless floor. Coefficient of friction between the bars is μ. Initially both the bars are moving towards a wall with a velocity u. The wall is parallel to the front face of the bars. If collision of the bars with the wall is perfectly elastic, find final velocities of both the bars long time after collisions with the wall.



Solution: Immediately after collision, situation appears as shown in adjacent figure. The speeds of the two blocks are equal and opposite and so also their decelerations so their speeds remain equal and opposite until (and if) the lower block also collides, after which the speeds of both blocks will be same and the slipping will stop. Now there are two possibilities:



① lower block also collides

② lower block does not collide.

For lower block to collide, it must move a distance l_0 relative to ground from the shown configuration and since upper block also has same speed ∴ upper block must also move l_0 in opposite direction. Thus relative slipping distance of one block on the other until collision becomes $2l_0$. In this process loss in mechanical energy is $\Delta Q = 2\mu mg l_0$.

If this is more than initial energy it would mean that blocks came to rest before second collision. Thus:

If $\frac{1}{2}mu^2 \times 2 < 2\mu mg l_0$, final speed will be zero i.e.

if $u < \sqrt{2gl_0}$, $v = 0$ Ans. (v: final speed), else by WET:

$$\frac{1}{2}mv^2 \times 2 = \frac{1}{2}mu^2 \times 2 - 2\mu mg l_0 \Rightarrow v = \sqrt{u^2 - 2\mu g l_0}$$ Ans.

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